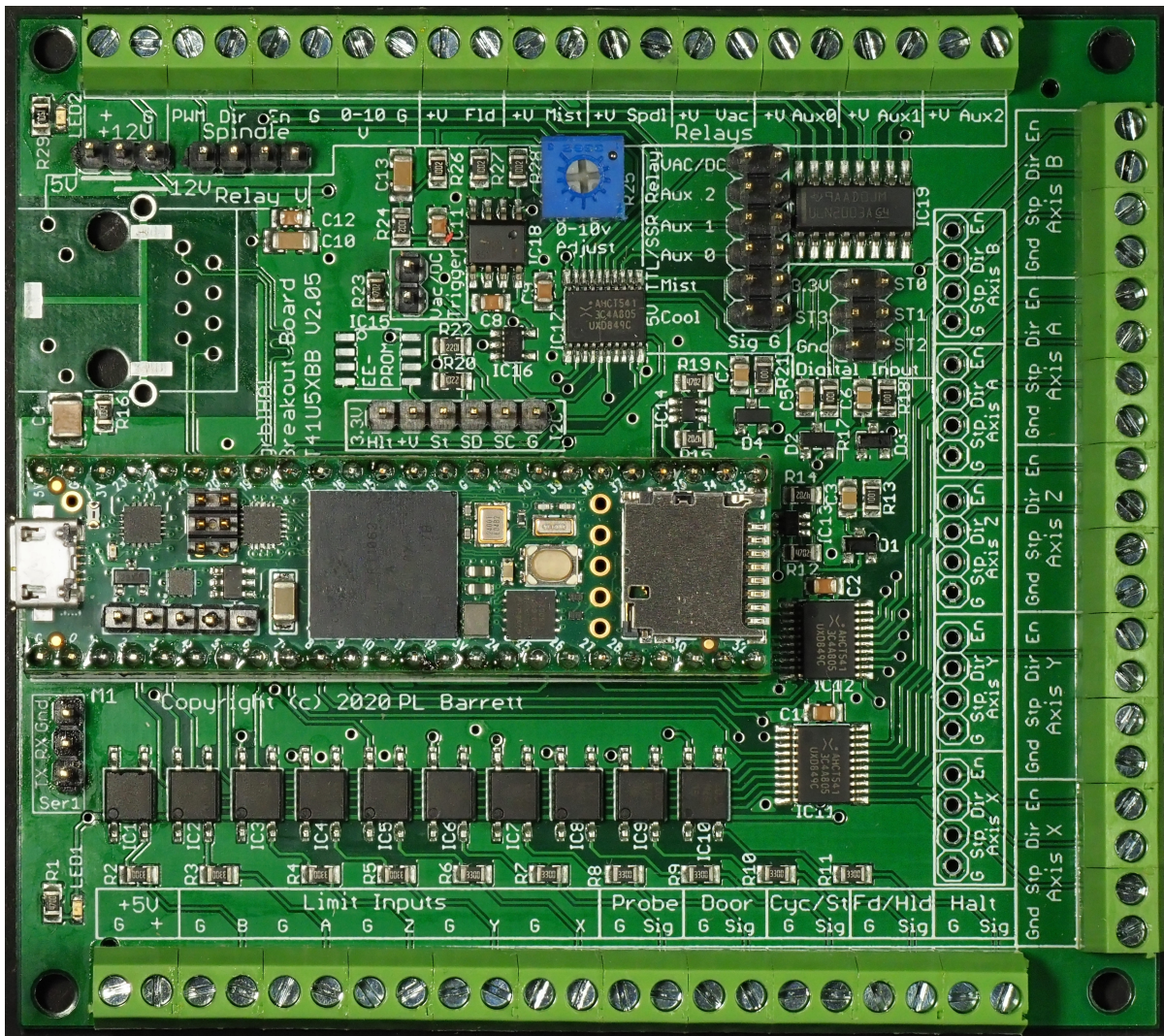
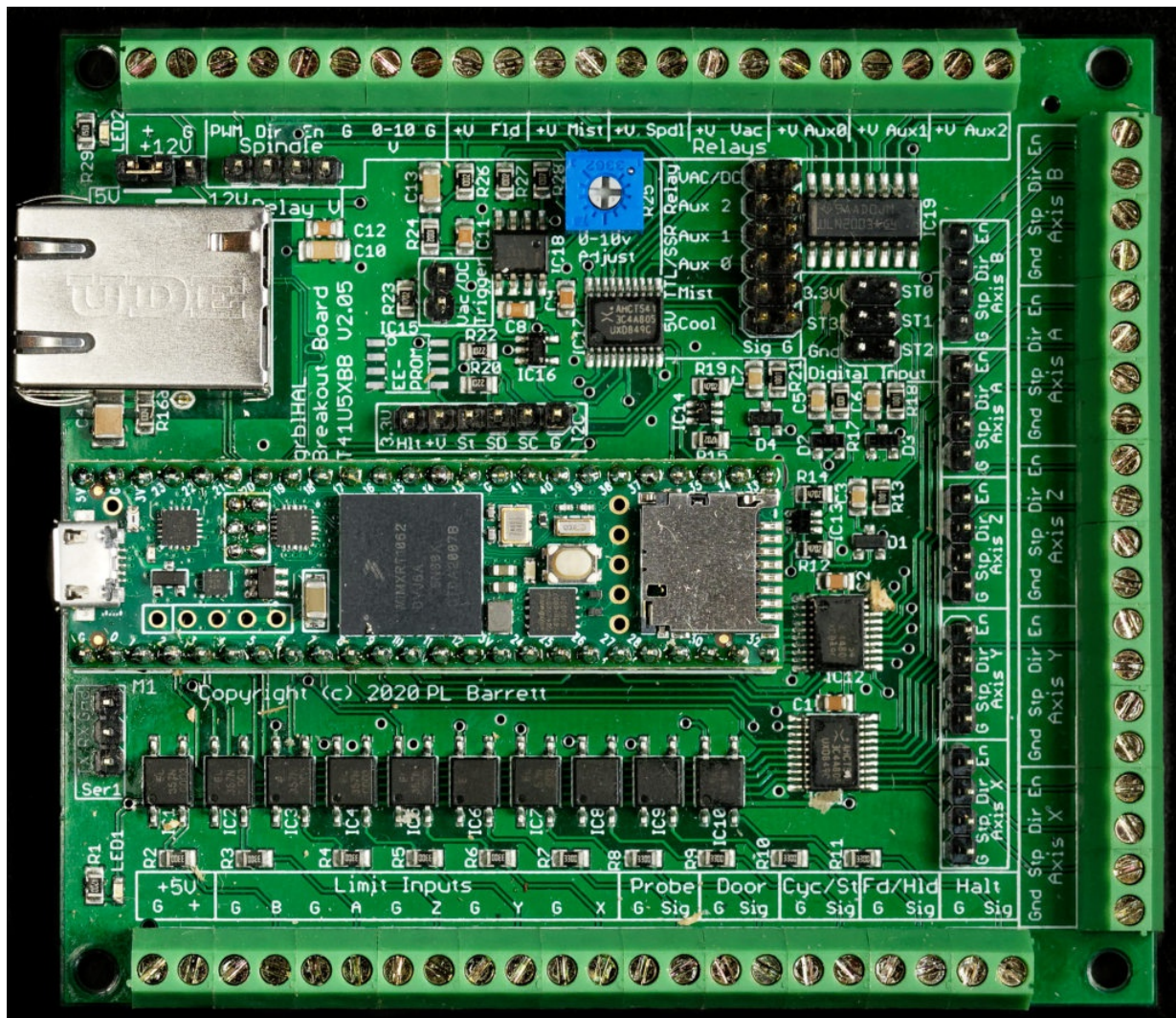


grblHAL Teensy 4.1 Motion Controller Breakout Board User Manual

Versions 2.07, 2.09, 2.10, 2.11



T41U5XBB – 5Axis, USB connected Breakout Board



T41E5XBB – Ethernet Connected Breakout Board
(note silkscreen says T41USB5XBB, both share a common PCB)

Features:

- 5 Axis control outputs – 5V compatible.
- Independent enables for each axis.
- 10 Opto-isolated inputs including standard grbl controls and limits switches for all axes.
- 7 Relay Outputs – Spindle, Mist Coolant, Flood Coolant, Dust Extraction and 3 auxiliary relays. Each output can control a relay coil directly or 5V TTL (SSR compatible) devices.
- Dust Extraction relay support. Activated by spindle enable.
- Separate control of Dust Collector relay via pin header.
- Relay coil voltage switchable between 5V and 12V.
- Standard GRBL spindle control outputs – 5V compatible.
- 0-10V spindle control output.
- Screw Terminals for reliable connections available for most I/O.

- 4 Digital Inputs – EMI protected, 15.9KHz low pass filter, Schmitt Trigger.
- I2C header.
- EEPROM footprint for SOIC8 devices.
- LED indicators for 5V and 12V.
- USB interface.
- Ethernet interface.
- Serial I/O Header

Stepper Driver outputs.

There are 5 sets of standard GRBL stepper driver outputs: X, Y, Z, A and B axes. Each has 3 pins: step, direction and enable. Each are 5V compatible and capable of driving opto-isolated stepper driver inputs. Each axis has a set of screw terminals as well as pin headers (0.1"/2.54 mm spacing).

Limit Pins.

There are 5 limit inputs: X, Y, Z, A and B. All are opto-isolated and available via screw terminals.

Control Input Pins.

Standard GRBL control input pins are supported via screw terminals: Feed/Hold, Cycle/Start, Halt/Stop and Safety Door open. These are opto-isolated.

Probe Input

Probe input is opto-isolated and available via screw terminals.

Spindle Control.

Outputs for Spindle Enable, Spindle Direction and PWM signals available via screw terminal and pin header. These are 5V TTL logic compatible. There are relay outputs for the spindle (direct relay coil and 5V TTL). See Relay section for more details. In addition, standard 0-10V output for spindle control is provided via a screw terminal.

Relay outputs.

Standard GRBL Flood and Mist relay outputs are supported via screw terminals. Spindle relay output via a screw terminal is supported and controlled by Spindle Enable. In addition, a separate screw terminal relay output for Dust Collection is activated by the Spindle Enable output.

3 auxiliary relay outputs are supported via screw terminals and used via M62-M65 grblHAL pin control.

All relay coil outputs are capable of directly driving relay coils. Each relay driver is capable of up to 100 mA. All 7 are limited to 500 mA in total. Above 250 mA, forced air cooling is recommended.

All relay coil drivers have open collector outputs.

Each relay also has a 5V TTL output that can be used to drive solid state relays, separate relay boards or other low voltage activated devices.

Relay coil voltage is selectable between 5 and 12V/24V via the pinheader labeled Relay Voltage in the upper left corner of the PCB or, alternatively, via a solder jumper on the bottom the PCB.

Schmitt Triggered digital inputs:

Four digital inputs are provided via a pin header. These have Shottky Diode protection and a low pass RC filter with an 15.9 KHz cutoff frequency. See the schematic for the specific circuit.

I2C Header

An I2C header is provided. 3.3V only, there is no translation to/from 5V and the inputs are not 5V tolerant. A strobe line from the Teensy 4.1 pin 33 is provided on the header. In addition, a line for an E-Stop signal is routed directly to the grbl Halt input. Currently, only a limited number of devices are supported. Check the grblHAL github site for more details.

Versions

There are two versions of this board. Currently, they share the same printed circuit board. However, in the future, that may change.

T41U5XBB – this is the USB interface version and is recognizable by the lack of the Ethernet connector. The controlling PC is connected to the T41U5XBB via the microUSB connector on the Teensy 4.1.

T41E5XBB – this is the Ethernet version. The controlling PC is connected the T41E5XBB via the Ethernet plug on the base PCB.

In both cases you need to load the appropriate version of grblHAL. A T41E5XBB may be used as a T41U5XBB by loading the USB version of grblHAL. See Loading Firmware section for details.

External Connections

Power

12V, Input

Supply for relays and 0-10V output amplifier.

5V Input or Output

Supply for 5V logic and Teensy 4.1 when not power via microUSB connector.

Can supply up to 150 mA to off board devices.

Stepper motor, Output

5V signals.

Pin	Signal
G	Ground
Stp	Step Pulse. Positive.
Dir	Direction
En	Enable

Limit Switches, Input

One input for each axis. Opto-isolated, 5V.

Halt, Door, Cycle/Start and Feed/Hold, Input

Opto-isolated, 5V.

Probe

Opto-isolated, 5V.

Spindle Group, Output.

PWM, Direction, Enable, Ground. This is a 5V group.

0-10V.

Relay Coil Group, Output

These are open collector sinks. They run at the selected relay voltage.

Flood, Mist, Spindle, Vac/Dust Collector, Aux0, Aux1, Aux2

Relay TTL Group, Output.

These are 5V, TTL compatible outputs suitable for SSR and powered relay modules.

Flood, Mist, Vac/Dust Collector, Aux0, Aux1, Aux2.

Note: Enable in the Spindle Group is capable of driver SSRs and powered relay modules.

Vac/DC trigger.

External Switch to activate a vacuum or dust collector.

Digital Input

4 input pins, 3.3V and Ground. Schmitt Triggered input, diode protected and low pass filtered. See schematic for more details.

Pin	Signal
St0	Schmitt Triggered, Low Pass Filtered (15.9KHz)
St1	Schmitt Triggered, Low Pass Filtered (15.9KHz)
St2	Schmitt Triggered, Low Pass Filtered (15.9KHz)
St3	Schmitt Triggered, Low Pass Filtered (15.9KHz)
3.3V	3.3VDC
Gnd	Ground

S0-S3 are 5V tolerant. Note that V2.05 is not 5V tolerant but V2.07 and later are.

I2C Header, Input/Output.

Pin	Signal
G	Ground
SC	I2C SCL
SD	I2C SDA
St	Strobe
+V	3.3VDC, source by Teensy 4.1
Hlt	Halt Signal. Connected to the Halt input Teensy 4.1 pin.

PCB Assembly

Not all components are required. See Optional Components section for more information.

Preparing the Teensy 4.1

See the blog post about [building the T41U5XBB in pictures](#).

Before installing components, you need to decide how you will connect the Teensy 4.1 to the breakout board. There are two approaches:

1. Use sockets to mount the Teensy to the breakout board. This is the preferred approach. It is more flexible but runs the risk of shaking loose in a moderate to high vibration environment.
2. Directly solder the Teensy to the breakout board. This is the recommended approach for high vibration environments. It makes for a much more secure connection. This is also easier than the socket approach if using Ethernet.

Materials for directly soldering. See BOM section for ordering numbers.

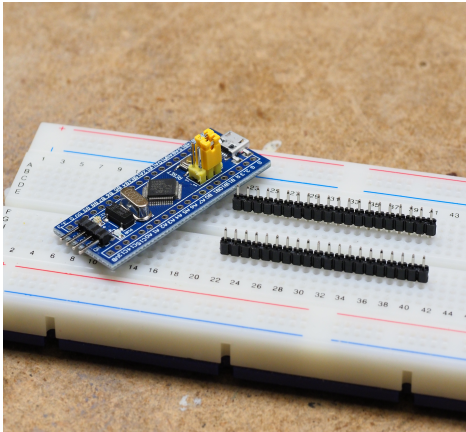
- 2 24 pin 0.1" pin-header strips. These can be obtained from numerous sources.
- 1 2x3 2 mm pin-header for the Ethernet connection.

Materials for socketing. See BOM section for ordering numbers.

- 2 24 pin 0.1" pin-header strips. These can be obtained from numerous sources.
- 2 24 pin 0.1" sockets.
- 1 riser 2x3 2 mm pin header. (For Ethernet)
- 1 2x3 2 mm socket. (For Ethernet)

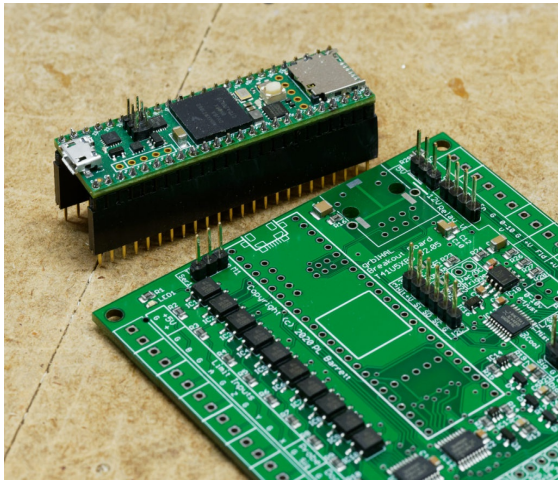
The T41U5XBB is provided with all surface mount components soldered in place. You are responsible for installing the through-hole connectors, either from the option assembly kit or sourced yourself. Assemble the PCB in the following order:

1. Install 1x2 and 1x3 pin headers: Vac/DC Trigger, 5/12V selection and Ser1. Carefully snap each one off the longer pin-header strip. Use tape (painter's tape works well) to hold them in place. Solder 1 pin and then verify the alignment. The header should be perpendicular to the board. If it is not, reheat the soldered pin and gently press it into place (watch the heat on the short headers!). Be sure to be gentle with heating headers as the plastic spacer softens quickly.
2. Install longer 1x headers : Spindle (1x4), I2C (1x6) and 5 Axis Headers (1x4). Use same technique as in 1).
3. Install 2X headers: TTL Relay Header (2x6) and Digital Input Header (2x3). Use same technique as in 1).
4. If you bought a Teensy 4.1 without headers, install pin headers. A solderless breadboard makes a great way to stabilize the pin headers while soldering the 0.1" header strips. (The photo below is just for illustration, that is not a Teensy.)



If using Ethernet, solder the 2x3 2 mm header in place on the bottom of the teensy 4.1. Use tape to hold it in place, solder 1 pin and check for alignment. If misaligned, reheat the soldered pin and press the header into correct alignment. Then, solder the other pins.

5. Install R25. For V2.09 and earlier only. V2.10 and later have R25 already installed so skip this step. You can bend the leads to ensure it stays in place. But, also follow the the above approach of soldering 1 pin first, verifying alignment and then finishing. Note that a dual pinout footprint is used to allow several different types of trimmer potentiometer to be used. See BOM for more detail.
6. Install the Teensy 4.1. If using the sockets approach, place the sockets on the pin headers you just soldered onto the Teensy 4.1. Note, in the picture below, the 2x3mm pin header on the Teensy 4.1 is for testing purposes and not needed for the USB version. The Ethernet version uses a 2x3mm header that is attached to the underside of the Teensy.



Now, gently inset the socketed Teensy 4.1 assembly into the indicated holes on the breakout board. Verify that all socket pins are correctly in place. Secure the entire assembly with tape on the component side of the board and then solder one pin on each of the different sockets and verify alignment. Once you are sure that the alignment is correct, solder the other pins, leaving the pins you initially soldered to last.

7. If using the direct solder approach, gently insert the Teensy 4.1 into the appropriate holes on the breakout board. Verify that all pins are correct and use tape on the component side to secure the Teensy 4.1 in place. Make sure the 2x3mm header for the Ethernet connection is properly inserted in the corresponding holes in the breakout board. Then solder 1 pin on each header, verify alignment and correct if necessary. Once satisfied with the alignment, solder the other pins, leaving the pins you initially soldered to last.
8. Mount and solder screw terminals. Assemble a complete row before soldering. The terminals lock together but are sometimes hard to get completely aligned. If that is the case, place them upside down (with the pins pointing up) on a hard surface and press each terminal in place with a screwdriver or similar. When you are satisfied that the tops are flush, insert and solder an end pin. Then solder the pin at the other end, making sure the row is properly aligned. It is sometimes helpful to solder the pin of middle a middle terminal while moving the row into proper alignment. Then solder remaining terminals while ensuring that the row is straight

Optional Components.

Many components are optional. Refer to the following diagram.

Notes:

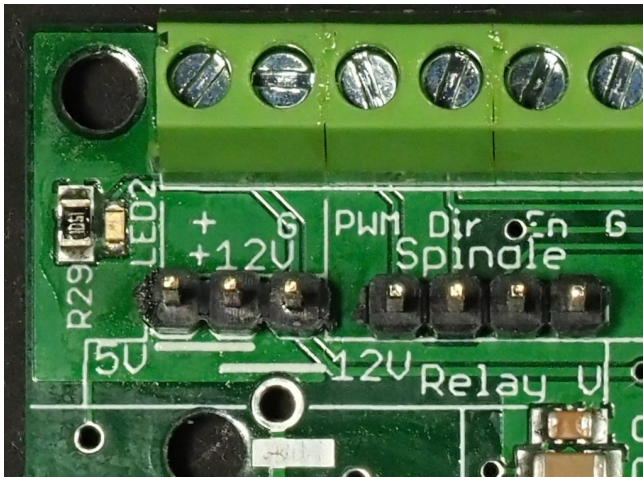
- Ser1 and I2C headers are optional. grblHAL will work correctly without either header installed. Specific options may need them.
- EEPROM is optional, Install if you need to use an external EEPROM. grblHAL will work correctly without it.
- 0-10V Spindle control is optional. Use if you are using a VFD controlled spindle. grblHAL will work correctly without it if you are manually controller your spindle. We recommend installing the screw terminals as they are difficult to add later.
- Relays are optional. The T41E5XBB and T41U5XBB is capable of driving relay coils directly. It is recommended that you install the screw terminals even if you don't initially plan on using them. Installing later can be difficult. See note on Vacuum/Dust Collector relay below. But, grblHAL will work correctly without any relays.
- A and/or B Axes are optional if you don't have them in your CNC machine.
- Safety Door alarm. Install if you have an enclosure for your CNC machine with a Safety Door sensor. grblHAL will work correctly without it but you will need to load the a version of grblHAL with it disabled. Alternatively, you can simply put a jumper in the Door terminal.
- Axis pin headers. There are 5 sets of pin headers, one for each Axis. Install these only if you intend on using a them to connect to your stepper drivers. For reliability, screw terminals are recommended over pin headers. If using pin headers, we recommend using hot glue to secure the connector in place.
- Ethernet MagJack. If you are not using Ethernet, there is no need to install this. Currently, both T41E5XBB and

T41U5XBB boards have the footprint for the MagJack. Future versions of the T41U5XBB may not have this footprint.

- In general, we recommend that you install all screw terminals at the beginning. Adding them later can be difficult due to their interlocking nature.

Relay Voltage

Relay Coil Voltage source is selectable between 5V and 12V (or higher) provided via an external power supply. Use a “suitcase jumper” as indicated on the PCB to select the desired voltage.

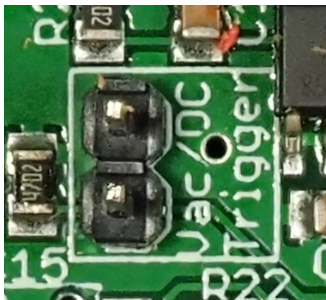


Note, we recommend you use an external 5V supply if you are driving 5V relay coils directly. To do so, you must disconnect the Teensy 4.1 +5V feed on the PCB. You will still need to power the Teensy 4.1 via the micro-USB connector. See Power Section for more information.

Relay coil voltage up to 24V is permissible and is applied at the input terminal labeled 12V. If using higher than 12V, we recommend changing resistor R29 to limit LED current to below 5 mA. For 24V, use 4.7K Ohm. The user is responsible for ensuring that all relays are rated for the voltage they provide.

Note: it is recommended to not use the stepper motor supply for the relay supply. It could be a source of EMI and cause erratic operation of the system.

DustCollector/Vacuum Control



An additional driver is provided for a dust collector relay. It is activated whenever the spindle is turned on (M3). In addition, this relay output can be independently activated by a switch that connects to the pin header labeled Vac/DC Trigger. This makes it easy to do clean up with the same Vacuum or Dust Collector used for dust extraction while running your CNC machine. Automatic triggering of the Dust Collector can be disabled, see the PCB Level Configuration Section.

Building and Loading Firmware.

The grblHAL Web Builder is the primary tool for building firmware for the T41U5XBB. It is available at <https://svn.io-engineering.com:8443/>. Select iMXRT 1062 (Teensy 4.1) as the Driver and T41U5XBB as the Board. In the General panel, select the Number of axes. If you have a dual motor moving gantry, select that axis as Ganged or Auto-squared. If you have installed the Ethernet kit, In the Network/WebUI panel, select Ethernet and check Telnet server and WebSocket server options. Press Generate and Download Firmware button and save the file for loading. Be sure to save the configuration by clicking on the Save Board button. This will allow to you to later make changes without having to remember what you selected.

Loading Firmware

You will use the teensy.exe application available from pjrc.com. Start with this page - <https://www.pjrc.com/teensy/loader.html> and follow the links for you specific operating system.

Building from source code

In general, building grblHAL from source code is well beyond the scope of this document. You can download the source code at <https://github.com/terjeio/grblHAL>. To build it, follow the instructions for Compiling GrblHAL in the grblHAL Wiki at <https://github.com/terjeio/grblHAL/wiki/Compiling-GrblHAL>. The grblHAL core Discussions section (<https://github.com/grblHAL/core/discussions>) is a good place to ask questions and get help.

Running grblHAL on your machine

Once you have finished assembly, your grblHAL Teensy 4.1 based Motion Controller is ready to be tried out. Attach it to a PC via a USB cable and run a compatible GRBL sender application on the PC. We recommend ioSender and use it in our examples: <https://github.com/terjeio/ioSender>. Check for the most recent release. An overview of ioSender is available here: <https://www.grbl.org/single-post/one-sender-to-rule-them-all>.

Next, you will need to change the settings to reflect your hardware. Actual grbl setting values are also shown. Follow these steps:

1. Hook up your control buttons: Feed Hold, Cycle Start and Rest/EStop.
2. Open the **Settings: Grbl** panel. You will see a list of parameters that control grblHAL.
3. Set the “Invert Step Enable” mask. Check all boxes. \$4=7 for a 3 Axis machine. \$4=15 for a 4 Axis machine and \$4=31 for a 5 Axis machine
4. Set “Invert Control Pin” mask. Invert Feed Hold, Cycle Start, Reset and Safety Door. \$14 = 15.
5. Save these parameter and return to the main tab (**Grbl** panel, you should see the DRO section).
6. Press the Red Reset button if and ALARM state is showing.
7. Open the **Jog** panel and click on any of the job buttons. The DRO should show movement. If you have motors hooked up, they should show movement.
8. If you have any control buttons attached, test them. It should show activity. Note the associated “LED” in the G Code Sender application when you press a button. If nothing happens, you should verify your connections.
9. Close the **Jog** panel, fill in 1000 in the Spindle RPM box and click on the CW “radio” button. The LED on the Teensy should light up. Click on off and the LED should go off.
10. If everything works to your satisfaction, you should now proceed to full testing of your machine.

You can also refer to the grblHAL wiki's First Run Grbl Settings section: <https://github.com/grblHAL/core/wiki/First-Run-Grbl-Settings>

Some Initial GRBL Settings

GRBL settings are dependent on your hardware so you might have to experiment a bit. Here are some key settings that will help you get going faster.

\$ID	Name/meaning	Suggested Value	Comments
0	Step Pulse time in uS	5	Default is 10 but 3 or 5 should work ok. Using the default may cause problems for step rates above 80 kHz.
4	Step Enable bit mask, one per axis	7, 15 or 31	A 1 bit means invert. A 3 axis machine would have 7 as the value, a 4 axis machine would have 15, a 5 Axis - 31. Bit 0 is X, 1 is Y, 2 is Z, A is 3 and B is 4.
14	Control pin inversion bit mask, one per pin.	30	A 1 bit means invert. Reset is bit 0, Feed Hold – 1, Cycle Start – 2, Safety Door – 3.
30	Max spindle speed in RPM	1000 or VFD max.	Default is 1000. If you have a VFD, you will need to set this to the spindle max. In G-Code, the Sxxx parameter controls speed and to get full speed, you will need have xxx match this parameter (S1000, in the case of the default).
100, 101, 102, 103, 104	Steps/mm. One for each Axis. 100 – X, 101 – Y, and so on.	varies	Set these based on your hardware – lead screw, belt ratios, microstepping. The default is likely wrong for your machine but you will still get movement. On a per axis basis, determine the number of steps (including microsteps) it will take to make one full rotation of the lead screw. Determine how many mm that axis will move with one full rotation of the lead screw. Divide the first number by the second to get the value you will use.

Note, if you don't attach any switches, the system may come up in the alarm state before you change any settings. This is normal. If you are not using the Safety Door Alarm, in the **Settings: Grbl** tab, **Invert control pins** section, check the **Safety Door** box. This makes the system see the (nonexistent) Safety Door as closed. Similarly, you can invert the Halt. However, we highly recommended that you do not disable the Halt function and that you install a dedicated Halt button in a prominent location on your CNC machine.

You should spend some time studying all the GRBL settings. We suggest you start conservatively and tune your machine to higher performance one parameter at a time.

Compatible G-Code Senders

Most senders will work with grblHAL. If your preferred sender is having problems, it may need to use a version of grblHAL built with compatibility settings. There are pre-built versions available, see the Loading Firmware section above.

Setting up Ethernet.

If you wish to use Ethernet to connect to your breakout board, you will need to solder the network interface connector MAGJack in place. It is available in the Ethernet kit. See the BoM section for the part number if you don't want to buy the Ethernet kit. Once you have soldered it on the PCB, you will need to load a build of grblHAL that has been configured to use networking. You will use the USB connection to do that.

You will next need to load an Ethernet build of the firmware. Available via the grblHAL Web Builder (<https://svn.io-engineering.com:8443/>). Optionally, you can build a version from the source code at the grblHAL Github site.

To set up Ethernet, refer to this blog post for instructions: <https://www.grbl.org/single-post/networking-the-t41u5xbb>

Direct PC Connection via Ethernet

Many CNC users like to use a direct Ethernet connection between the PC and CNC machine. The advantage of this is the CNC machine is not visible from your network and is thus safe from intrusion and interference. Because grblHAL is not a secure system, this is the only way to network secure your CNC machine. Refer to this blog post for instructions: <https://www.grbl.org/single-post/ethernet-without-routers-for-grblhal>

Note that modern network switches prevent most interference and are quite inexpensive.

0-10V output setup

The 0-10V circuit works by using an opamp to filter a 5V PWM signal and amplify it to 10V. As such it needs 12V or higher input which the user must provide via the 12V screw terminals. A second order filter with an aggressive low pass cut off frequency is used to provide a stable signal for the spindle controller.

Because source voltage level can vary, the amplifier must be adjusted. You can do this by turning on your spindle in GCode (via Sxxxx) to full speed and adjusting R25 until the voltage on the 0-10V terminal reads 10V. Here is a step-by-step method you can use:

- Disconnect your spindle or VFD.
- In the **Settings: Grbl** tab, set \$30, **Maximum Spindle Speed**, to 1000.
- Send the following command to grblHAL: **M3 S1000**. This turns on the spindle and causes the speed to be set to 100%. The LED on the Teensy 4.1 should light up.
- Measure the 0-10V output and G terminals.
- With a screwdriver, adjust the potentiometer labeled “0-10V Adjust” until the voltage is as close to 10V as you can get. Anything in the range 9.990V to 10.010V is acceptable.
- Send the following command to grblHAL: **M5**. This turns off the spindle.
- Reconnect your spindle or VFD and test.
- You may want to change the max spindle speed setting, \$30, to something more appropriate for your machine.

You may want to repeat the above process with the spindle connected and running to fine tune the maximum speed.

PCB Level Configuration

A number of features can be configured via solder jumpers on the bottom of the PCB.

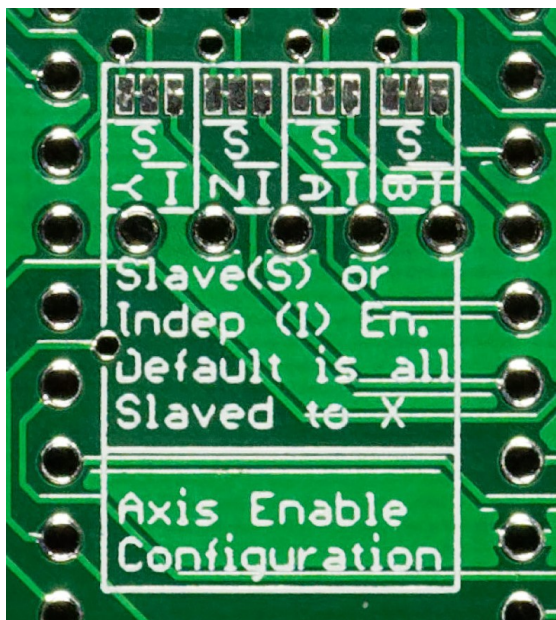
Notes:

1. To help seeing any trace you need to cut, carefully scrape the green solder mask off the area between the pads. The trace should show up as bright copper. This will also make a solder bridge easier to create.
2. When you cut a trace, be sure to clean out and inspect the cut to ensure no stray slivers of copper are causing a short.

Individual Stepper Enables

No longer available as of V2.10.

By default, all 5 Axis Stepper enable signals are tied together. This is a common CNC machine configuration. If your machine needs separate enables for all or specific axes, you can configure this in the Axis Enable Configuration block on the bottom of the PCB.



To make a given axis have an independent enable signal:

1. Locate the 2 way solder jumper associated with the axis you want to make independent. Note which side has the connection. It is marked with a horizontal line and an S.
2. Using a sharp knife, cut the trace between those two pads.
3. With a soldering iron, tin the center pad and the one indicated with an I.
4. Then slowly add additional solder to the space between the I and center pads until there is a bead of solder that bridges between them.

If you wish to undo the configuration at a later date, use a soldering iron and wick to remove the solder bridge and then create a new one between the center and S pads.

External 5V Feed

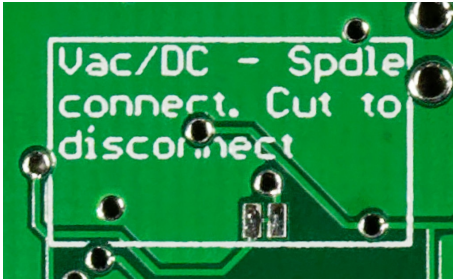
The Teensy 4.1, when connected via USB, normally has enough power to support the 5V needs of the breakout board. You may wish to use external 5V in certain cases. We recommend you do this if using 5V relays. In that case, you will need to disconnect the base PCB from the Teensy 4.1's 5 volt supply. Note that there are 2 traces connecting the two pads. Disconnect 5V by cutting both traces between the pads in the external 5V block on the bottom of the PCB.



Using a sharp knife carefully cut the two traces. Use the tip of the knife to ensure there is no connection. Test it for continuity with a multi-meter. When powered up via USB only (i.e. no additional 5V source used), the LED next to the +5V screw terminal should not light up. If it does, you need to repeat the above process. You will need to separately supply power to the Teensy 4.1, via USB from a computer or charger.

This process may be reversed by bridging the two pads with solder as described the above section.

Vacuum/Dust Collector relay



By default, the Vacuum/DC Relay is activated when the Spindle is enabled. You may disable this by cutting the trace between the two pads in the Vac/DC – Spindle Connect block on the bottom of the PCB. No longer available as of V2.10.

This may be reversed by bridging the two pads with solder as described the above section.

Link A Axis to Y

No longer available as of V2.10.



This is for a dual motor Y axis configuration. The default configuration is A is not linked to Y. When jumpered to Y, both Y and A axis motors will run in tandem, both getting the same signals. Using a sharp knife, carefully cut fully between the pads marked A (right two pads) and use solder to short the pads marked Y (left two pads). You should verify that there is no continuity between the two A axis pads as that can cause board damage.

Note that the traces to be cut may not be exactly centered in the pads.

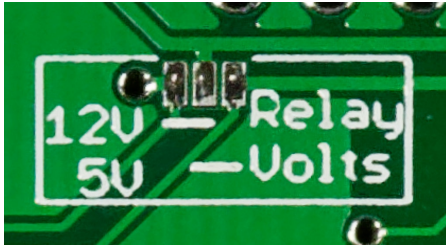
This process may be reversed by bridging the two pads with solder as described the above section.

Note: this is not needed for current grblHAL and will not support Auto-Squaring. Configure Y as an Auto-squared Axis in the Web Builder.

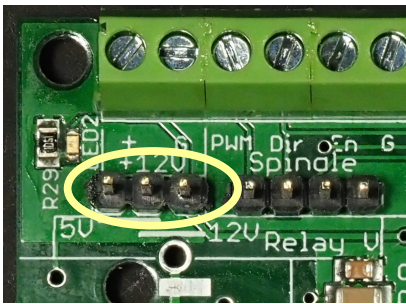
Relay Coil Voltage

No longer available as of V2.10.

Similar to the pin header selection of the relay coil voltage, you may configure the relay voltage via solder jumpers.



The default configuration has neither connected. Use the above solder jumper process to bridge the appropriate side of the two way jumper. You will need to short either the left pair of pins together for 12V or the right pair together for 5V.



WARNING. If you do this, it is highly recommended that you remove (or not install) the pin headers from the relay voltage selection section to prevent accidentally routing 12V into the 5V sections of the board. Failure to follow this precaution will cause you to destroy your breakout board and the Teensy 4.1 by inadvertently shorting 5V to 12V.

Power and EMI Considerations

Clean, EMI-free power is important for smooth operation of the grblHAL motion controller system. In designing the T41E5XBB and T41U5XBB, we used good practices to avoid EMI but further precautions are recommended. Power is often a source of EMI and you should take additional steps to ensure that these sources are managed. Finding the source of interference later can be difficult – it is good practice to design your system with avoiding it in mind. General recommendations:

- Do not power 5V relay coils from the USB voltage supplied by the Teensy 4.1. Relays can produce EMI and power fluctuations. Use external power (see above).
- Do not power relay coils directly from the Stepper Motor power supply. It can be a significant source of EMI.
- Use shielded wires for signals. Make sure the shields are grounded in a star configuration at the controller box or other common ground.
- Use ferrite filters on all wires entering or exiting the T41U5XBB or T41E5XBB PCB.
- Avoid routing wires near EMI sources such as a VFD, Stepper drivers, or power supplies. Avoid co-routing with wires that are connected to those devices as well.
- If you are connecting to the Teensy 4.1 via a USB cable, use one with a ferrite filter on it or add a clamp-on one close to the Teensy end of the cable.

If you are using Ethernet, and expect to not connect the Teensy 4.1 to a PC via USB, you will need to provide 5V for the Teensy. There are two ways to do this:

- Use a standard USB “charger” to power the Teensy 4.1 via the microUSB connector. We suggest a 1 Amp minimum capacity. This will work well if you aren't using 5V for relay coils.
- Provide 5V via the +5V screw terminal.

We recommend that all stranded wires have ferrules crimped on the end that goes into a screw terminal. This is a good practice in that it ensures the best possible contact and helps to avoid stray “cat whisker” strands of wire that could cause errors.

Using USB voltage for PWM control of VFDs

In some cases, the voltage supplied by USB may be lower than 5V. While the specification defines 4.75V as the minimum, some laptops/computers USB ports output even lower voltage. This can cause lower than requested speeds when using the direct PWM signal to control a VFD/Spindle. This does not effect the 0-10V output. This problem can be solved 2 ways:

1. By using external +5V to power the board. Remember to follow the instructions in the above “External 5V Feed” section.
2. By using the 0-10V output signal connected to the 0-10V input of the VFD. This requires you provide 12V to the board.

Errata:

1. V2.05 digital inputs ST0-ST3 are not 5V tolerant. They will work correctly at 3.3V. This is corrected in V2.07.
2. V2.05 silk screen labels the Flood connection as Cool. This is corrected in V2.07.
3. V2.10 silk screen labels for Aux0, Aux1 and Aux2 are slightly misaligned. Corrected in V2.11.

Schematics:

Refer to the github repository for current versions schematics.

Bill of Materials for through hole components (for the “Unkit” version).

Item	Number	Source
2 pin 3.5 mm Screw terminal.	32	Readily available on ebay/amazon/aliexpress/...
1x2 2.54 mm pin header	1	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [2]
1x3 2.54 mm pin header	2	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [2]
1x4 2.54 mm pin header	6	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [2]
1x6 2.54 mm pin header	1	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [2]
2x3 2.54 mm pin header	1	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [2]
2x6 2.54 mm pin header	1	Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [2]
1x24 2.54 mm pin header	2	For Teensy 4.1. If you purchase a Teensy with pins, omit these items. Readily available on ebay/amazon/aliexpress/... Snap off of longer strips. [2]
1x24 2.54 mm socket	2	Readily available on ebay/amazon/aliexpress/... [2][3] Higher Quality: Mouser – 801-87-024-10-001101, Digikey - 1212-1209-ND
Teensy 4.1	1	Available from pjrc.com.
R25, 10K Potentiometer	1	Numerous sources. Mouser: 652-3362P-1-103LF, Digikey: 3362P-103LF-ND [1] [5]
T41E5XBB (Ethernet)		
MagJack	1	Whiznet RB1-125BAG1A Mouser: 950-RB1-125BAG1A Digikey:1278-1011-ND
2x3 2 mm Header	1	Digikey:MTMM-103-10-T-D-355-ND, Mouser: 200-MTMM10310TD355 (drop ship from Samtec)[4]
2x3 2 mm Socket	1	Digikey: 17-2314820-6-ND, Mouser: 855-M22-7140342[4]

Notes:

1. The PCB has a dual footprint for the 10K trimmer potentiometer, R25. and will accept 2 different footprints: 3 pins in a row with 0.1”/2.54 mm pitch or a triangular layout with the middle pin offset by 2.54 mm. The first footprint is used by more expensive, higher quality multiturn trimmers and the second is very common and low cost. A multiturn trimmer will give a finer adjustment of the 0-10V output level. The BOM has the lower cost version. Also, you could use a 2K or 5K trimmer in place of the 10K one.
2. We recommend gold plated pin headers for reliability.
3. Single row sockets are also readily available from numerous sources though good quality ones are not that expensive. We recommend you use good quality, gold plated headers and sockets. Note: for using Ethernet with a socketed Teensy 4.1, care must be taken to select 2 mm and 2.54 mm headers and sockets that are similar in height. However, we recommend that you directly solder your Teensy 4.1 to the T41E5XBB PCB if using Ethernet.
4. Use if you decide to socket your Teensy 4.1. The 2 mm header and socket must be matched to the Teensy 4.1 socket and header height. If you solder your Teensy 4.1 in place, you will only need a 2x3 2 mm header but need to be sure it will be tall enough.
5. As of V2.10, this is included on the base board and is no longer part of the Through Hole BOM.

BOM for surface mount components.

Qty	Designator	Footprint	Comment
6	C1, C2, C8, C9, C11, C12	C0805	100NF 50V 0805 ROHS
2	C10, C13	C1206	1UF 50V 1206 ROHS
4	C3, C5, C6, C7	C0805	10NF 50V 0805 ROHS
1	C4	C1210	1NF ±10% 2KV X7R 1210
4	D1, D2, D3, D4	SOT-23-3	BAT54S
10	IC1, IC2, IC3, IC4, IC5, IC6, IC7, IC8, IC9, IC10	SOP-4_3.6x4.4x2.54P	EL357N(C)(TA)-G
3	IC11, IC12, IC17	TSSOP-20_4.4x6.5x0.65P	74AHCT541PW,118
2	IC13, IC14	SOT-363	74LVC2G17GW,125
1	IC16	SOT-753	SN74LVC1G32DBVR
1	IC18	SOIC-8_3.9x4.9x1.27P	LM358DR2G
1	IC19	SOIC-16_3.9x9.9x1.27P	ULN2003ADR
2	LED1, LED2	LED_0603	19-217/GHC-YR1S2/3T
1	R1	R0805	680OHMS ±1% 1/8W 0805 ROHS
4	R12, R14, R15, R19	R0805	47KOHMS ±1% 1/8W 0805 ROHS
1	R16	R0805	1MOHMS ±1% 1/8W 0805 ROHS
2	R20, R22, R29	R0805	2.2KOHMS ±1% 1/8W 0805 ROHS
1	R23	R0805	100KOHMS ±1% 1/8W 0805 ROHS
10	R2, R3, R4, R5, R6, R7, R8, R9, R10, R11	R0805	330OHMS ±1% 1/8W 0805 ROHS
4	R24, R26, R27, R28	R0805	10KOHMS ±1% 1/8W 0805 ROHS
1	R13, R17, R18, R21	R0805	1KOHMS ±1% 1/8W 0805 ROHS